

Final overview

- Course utilizes calculus and linear algebra to form a basis of mathematical computations
- Other core math subjects like pre-calculus and algebra are touched on
- Quantifying Error: static and sequential errors
- Fixed point iteration: existence, uniqueness, and convergence
- Interpolation: polynomials, basis and coefficients, divided differences, degrees of freedom and oscillations
- Numerical differentiation: grid spacing, and location of derivative approximation
- Numerical integration: integral mean value, error analysis, extrapolation
- Numerical linear algebra: vector/matrix norms, convergence, special structures, decomposition, algorithms for special structures.

Review (Section 1.1)

Remark: The utility of continuous functions is the ability to pass the limit through the function.

$$\lim_{x_n \rightarrow x} f(x_n) = f\left(\lim_{x_n \rightarrow x} x_n\right) = f(x)$$

An aside on limits. Given functions $f, g \in C$ and $c \in \mathbb{R}$, we have the following rules as $x_n \rightarrow x$:

- $\lim_{x_n \rightarrow x} (f(x_n) + g(x_n)) = \lim_{x_n \rightarrow x} f(x_n) + \lim_{x_n \rightarrow x} g(x_n)$
- $\lim_{x_n \rightarrow x} (f(x_n) \cdot g(x_n)) = \left(\lim_{x_n \rightarrow x} f(x_n)\right) \cdot \left(\lim_{x_n \rightarrow x} g(x_n)\right)$
- $\lim_{x_n \rightarrow x} (c f(x_n)) = c \cdot \lim_{x_n \rightarrow x} f(x_n)$
- If $g(x_n) \in \text{dom}(f)$, then $\lim_{x_n \rightarrow x} f(g(x_n)) = f\left(\lim_{x_n \rightarrow x} g(x_n)\right)$
- $\lim_{x_n \rightarrow x} \frac{f(x_n)}{g(x_n)}$ is defined when $\lim_{x_n \rightarrow x} g(x_n) \neq 0$

Notice: The case $\lim_{x_n \rightarrow x} g(x_n) = 0$ is indeterminate and must be handled separately.

Common strategies include:

1. Algebraic simplification
2. Squeeze Theorem
3. Asymptotic analysis
4. L'Hôpital's Rule

Theorem (Cauchy Mean Value Theorem). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Assume further that $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof. Define the auxiliary function

$$h(x) = f(x)(g(b) - g(a)) - g(x)(f(b) - f(a)).$$

Note that h is continuous on $[a, b]$ and differentiable on (a, b) since f, g are. Also observe:

$$h(a) = f(a)(g(b) - g(a)) - g(a)(f(b) - f(a)),$$

$$h(b) = f(b)(g(b) - g(a)) - g(b)(f(b) - f(a)).$$

Expanding both and simplifying shows $h(a) = h(b)$.

By Rolle's Theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$. Now compute

$$h'(x) = f'(x)(g(b) - g(a)) - g'(x)(f(b) - f(a)).$$

Thus

$$0 = h'(c) = f'(c)(g(b) - g(a)) - g'(c)(f(b) - f(a)).$$

Rearranging yields

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

□

Theorem (L'Hôpital's Rule, 0/0 case). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) and continuous on $[a, b]$. Suppose

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0,$$

and that $g'(x) \neq 0$ for all $x \in (a, b)$. If the limit

$$\lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

exists (finite or infinite), then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

Proof. Fix $x \in (a, b)$. Apply the Cauchy Mean Value Theorem to f and g on the interval $[a, x]$. There exists $c \in (a, x)$ such that

$$\frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Since $\lim_{t \rightarrow a^+} f(t) = \lim_{t \rightarrow a^+} g(t) = 0$, we have $f(a) = g(a) = 0$. Thus

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}, \quad c \in (a, x).$$

Now take the limit as $x \rightarrow a^+$. As $x \rightarrow a^+$, we also have $c \rightarrow a^+$, so

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = \lim_{c \rightarrow a^+} \frac{f'(c)}{g'(c)} = L.$$

This proves the rule. □

Theorem. If f is differentiable at $a \in \mathbb{R}$, then f is continuous at a .

Proof. We must show

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Consider the difference:

$$\lim_{x \rightarrow a} (f(x) - f(a)).$$

Write this as

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot (x - a).$$

As $x \rightarrow a$, both numerator and denominator of the fraction go to 0, i.e.

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

is in a $\frac{0}{0}$ indeterminate form.

By definition of differentiability,

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a).$$

But we can also view this as a case for **L'Hôpital's Rule**, applied to

$$F(x) = f(x) - f(a), \quad G(x) = x - a.$$

Both $F(a) = 0$ and $G(a) = 0$, so by L'Hôpital:

$$\lim_{x \rightarrow a} \frac{F(x)}{G(x)} = \lim_{x \rightarrow a} \frac{F'(x)}{G'(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{1} = f'(a).$$

Therefore

$$\lim_{x \rightarrow a} (f(x) - f(a)) = \left(\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \right) \cdot \lim_{x \rightarrow a} (x - a).$$

The first limit is $f'(a)$ (finite by assumption). The second limit is 0. Hence the product is $f'(a) \cdot 0 = 0$.

Thus

$$\lim_{x \rightarrow a} f(x) = f(a),$$

which proves continuity. □

Definition (Uniform Continuity). A function $f : A \rightarrow \mathbb{R}$ (with $A \subseteq \mathbb{R}$) is called **uniformly continuous** on A if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all $x, y \in A$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Remark. - The key difference from pointwise continuity is that in uniform continuity the δ depends only on ε (and the function), *not* on the particular point $x \in A$. - In other words, a single δ works uniformly across the whole domain.

Example. The function $f(x) = x^2$ is not uniformly continuous on the real numbers as the function tends to ∞ . However, on a bounded interval it is.

Theorem. Any continuous function on a closed bounded interval is uniformly continuous.

Definition (Partition of an interval). Let $[a, b] \subset \mathbb{R}$ be a closed interval. A **partition** P of $[a, b]$ is a finite ordered set of points

$$P = \{x_0, x_1, \dots, x_n\}, \quad a = x_0 < x_1 < \dots < x_n = b.$$

Subintervals. Each consecutive pair (x_{i-1}, x_i) is called a **subinterval**, denoted

$$[x_{i-1}, x_i], \quad i = 1, \dots, n.$$

Mesh (Norm of the Partition). The **mesh** (or norm) of the partition P is

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

Tags. A **tagged partition** is a partition $P = \{x_0, \dots, x_n\}$ together with a choice of sample point

$$t_i \in [x_{i-1}, x_i] \quad \text{for each } i = 1, \dots, n.$$

The points t_i are called **tags**.

Definition (Riemann sum). Given a tagged partition $\{([x_{i-1}, x_i], t_i)\}$ of $[a, b]$, the **Riemann sum** of $f : [a, b] \rightarrow \mathbb{R}$ is

$$S(f, P) = \sum_{i=1}^n f(t_i) (x_i - x_{i-1}).$$

Definition (Riemann integral). We say f is Riemann integrable on $[a, b]$ if there exists a real number I such that for every $\varepsilon > 0$, there exists $\delta > 0$ with the property: for any tagged partition P with $\|P\| < \delta$,

$$|S(f, P) - I| < \varepsilon.$$

In this case we write

$$I = \int_a^b f(x) dx.$$

Equally spaced partition. For $n \in \mathbb{N}$, define the partition

$$x_i = a + i\Delta x, \quad \Delta x = \frac{b-a}{n}, \quad i = 0, 1, \dots, n.$$

Thus all subintervals have equal length Δx and the mesh is $\|P\| = \Delta x$.

Choose tags as the left endpoints $t_i = x_{i-1}$. The Riemann sum becomes

$$S_n = \sum_{i=1}^n f(x_{i-1}) \Delta x.$$

Convergence to the Riemann integral. If f is continuous on $[a, b]$, then f is uniformly continuous (by the Heine–Cantor theorem). Thus for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $|x - y| < \delta$, we have $|f(x) - f(y)| < \varepsilon$.

Now for the equally spaced partition with $\Delta x < \delta$, we know $|x - y| < \delta$ within each subinterval. This ensures that the variation of f on each subinterval is less than ε . Hence the difference between any tagged Riemann sum (e.g. left endpoints, right endpoints, or midpoints) and the integral is bounded by a term that goes to 0 as $n \rightarrow \infty$.

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_{i-1}) \Delta x = \int_a^b f(x) dx.$$

Errors (Section 1.2)

Motivation: What is the *computer representation of a number*?

Number systems

Defn: A *base* is the number used as the reference frame for constructing a number system.

e.g. Base-10: 0, 1, 2, 3, ..., 9

$$\text{ex: } 5239 = 5 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 9 \times 10^0$$

e.g. Base-2 (*binary*): 0, 1

$$\text{ex: } (1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

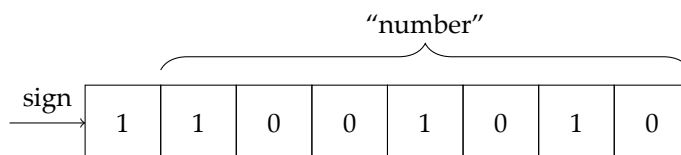
Each binary digit is referred to as a *bit*.

Binary representation of integers

Suppose we have 8 bits to represent an integer $z \in \mathbb{Z}$.

sign bit = 0 $\rightarrow$ positive integer

sign bit = 1 $\rightarrow$ negative integer



$$\text{Ex: } (11001010)_2 = (-1) \times (1 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0) = -(74)_{10}$$

Notice: the 8 bit representation has a range of $(-127, 127)$.

Prove by induction: the n bit representation has a maximum positive value of $2^{n-1} - 1$.

Binary representation of floating point numbers

Motivation: What if we want to include decimal representations?

$$\text{e.g. } (62.54)_{10} = 6 \times 10^1 + 2 \times 10^0 + 5 \times 10^{-1} + 4 \times 10^{-2}$$

$$\text{e.g. } (11.01)_2 = 1 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} = 3.25$$

Canonical Form

Notice that the decimal can be placed anywhere $(62.54)_{10} = 0.6254 \times 10^2 = 624500 \times 10^{-4}$

So we introduce a *canonical form*: 6.254×10^{-1} – one non-zero entry left of the decimal. (i.e. *scientific notation*)

General Form: In general, any $x \in \mathbb{R}$ can be written as

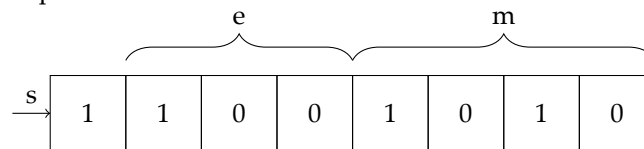
$$x = -1^s \cdot m \cdot b^e$$

where

- s : sign
- m : mantissa
- b : base
- e : exponent

Using the canonical form we take m to be normalized as $1.f$ (e.g. $*. * \dots \times 2^e$ where $*$ is a non-zero value)

Using 8-bit floating point representation



$$\text{Ex: } (11001010)_{\text{8-bit float}} = (-)(1.1010 \times 2^4) = (-)(1 \times 2^4 + 1 \times 2^3 + 1 \times 2^1) = -(26)_{10}$$

Floating point errors

Motivation: Machines represent $x \in \mathbb{R}$ within a certain level of precision using floating point numbers.

$$\text{True Value} = \text{Approximate Value} + \text{Error}$$

Defn: Machine precision, $\varepsilon_{\text{mach}}$ is the smallest $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$ such that $\text{fl}(1 + \varepsilon) > \text{fl}(1)$ where fl denotes the floating point representation of the number.

Floating point operations (flops)

Stability

Let $\mathbf{e}^0$ denote the initial error and $\mathbf{e}^n$ the error after n steps. A scheme is *stable* if there exists a constant C independent of n and of the input data such that

$$\|\mathbf{e}^n\| \leq C \|\mathbf{e}^0\| \quad \text{for all } n,$$

where h is the step size (or, more generally, a discretization parameter). This is *uniform boundedness* of the error amplification.

Remark: More generally, many error analyses yield

$$\|\mathbf{e}^n\| \leq C(n) \|\mathbf{e}^0\|,$$

for some function $C(n)$. Typical regimes:

- $C(n) \equiv C$ (uniform bound): errors remain uniformly controlled (unconditional stability on the given horizon).
- $C(n) = C^n$ with $C < 1$: geometric decay; with $C = 1$: bounded; with $C > 1$: geometric growth.

Remark (loss of uniformity). If the bound depends on discretization parameters (e.g., $C = C(h)$) or grows with n unless a step-size condition holds, then stability is *conditional*: the method is stable only when those conditions (CFL-type, spectral, etc.) are enforced. If no such bound exists, the method is unstable.

Linear iteration example. Consider the linear update

$$\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n, \quad \tilde{\mathbf{x}}_0 = \mathbf{x}_0 + \delta,$$

so the error satisfies $\mathbf{e}^n = \mathbf{A}^n \delta$ and

$$\|\mathbf{e}^n\| \leq \|\mathbf{A}^n\| \|\delta\|.$$

From submultiplicativity, $\|\mathbf{A}^n\| \leq \|\mathbf{A}\|^n$, hence

$$\|\mathbf{e}^n\| \leq \|\mathbf{A}\|^n \|\delta\|.$$

Consequences (in any induced operator norm):

- If $\|\mathbf{A}\| < 1$, then $\|\mathbf{e}^n\| \leq \|\mathbf{A}\|^n \|\delta\| \rightarrow 0$ (geometric decay).
- If $\|\mathbf{A}\| = 1$, then $\|\mathbf{e}^n\| \leq \|\delta\|$ (uniformly bounded).
- If $\|\mathbf{A}\| > 1$, the bound grows like $\|\mathbf{A}\|^n$ (at best), so errors can amplify.

A sharper spectral statement: $\mathbf{A}^n \rightarrow \mathbf{0}$ iff the spectral radius $\rho(\mathbf{A}) < 1$. For *normal* $\mathbf{A}$ (e.g., symmetric), $\|\mathbf{A}\|_2 = \rho(\mathbf{A})$, and the norm and spectral criteria coincide.

Takeaway. The goal of this class will be to analyze: (i) the error $\mathbf{e}^n$ and its amplification, and (ii) the bounds C (or $C(n)$). Uniform bounds yield unconditional stability; bounds that require parameter relations yield conditional stability.

Section 1.3 Algorithms and Convergence (Text Only)

Section 2.1 Root Finding

How to guarantee a solution exists? i.e. how to know $\exists x \in \text{dom}(f)$ such that $f(x) = 0$.

Due to the intermediate value theorem.

Note: IVT is proved by taking $s = \{x | f(x) \leq 0\}$ then for $f \in C[a, b]$, at least $a \in s$. But because f is continuous then for some $\epsilon > 0$ then $f(a + \epsilon) \in s$ as long as $a \neq \sup(s)$. Thus $c = \sup(s)$.

How to guarantee a unique solution exists? $\exists! x \in \text{dom}(f)$ such that $f(x) = 0$? – use the derivative.

Not universal but one common property is monotonicity $f \in C^1[a, b]$ and $f'(x) > 0$ for all $x \in [a, b]$.

Bisection Algorithm

Setup. Let f be continuous on $[a, b]$ with $f(a)f(b) < 0$. (Assume a unique root $p \in (a, b)$.)

Iteration. Initialize $a_1 = a$, $b_1 = b$. For $n \geq 1$:

$$p_n = a_n + \frac{b_n - a_n}{2} = \frac{a_n + b_n}{2}.$$

(note the first is expression more numerically stable).

$$\begin{cases} \text{if } f(p_n) = 0 \text{ then stop (root found);} \\ \text{if } f(a_n)f(p_n) < 0 \Rightarrow a_{n+1} = a_n, b_{n+1} = p_n; \\ \text{else } a_{n+1} = p_n, b_{n+1} = b_n. \end{cases}$$

Stopping criteria. For a tolerance $\varepsilon > 0$, terminate at the smallest N such that

$$\begin{aligned} |p_N - p_{N-1}| &< \varepsilon, \\ \frac{|p_N - p_{N-1}|}{|p_N|} &< \varepsilon \quad (p_N \neq 0), \\ |f(p_N)| &< \varepsilon. \end{aligned}$$

Analyzing the Bisection Algorithm

Theorem 1. Suppose $f \in C[a, b]$ and $f(a)f(b) < 0$. The Bisection method generates a sequence $\{p_n\}_{n=1}^\infty$ approximating a zero p of f with

$$|p_n - p| \leq \frac{b-a}{2^n}, \quad n \geq 1.$$

Proof. For each $n \geq 1$, we have

$$b_n - a_n = \frac{1}{2^{n-1}}(b-a)$$

and

$$p \in (a_n, b_n).$$

Since $p_n = \frac{1}{2}(a_n + b_n)$ for all $n \geq 1$, it follows that

$$|p_n - p| \leq \frac{1}{2}(b_n - a_n) = \frac{b-a}{2^n}.$$

□

Note: this is only a bound for the error and may be quite a conservative bound.

Section 2.2 Fixed Point Iteration

Definition 1 (Fixed point). A number p is a fixed point of a function g if $g(p) = p$.

Equivalence with root-finding. Root-finding $f(p) = 0$ and fixed-point problems $g(p) = p$ are interchangeable:

- Given $f(p) = 0$, define g so that p is a fixed point, e.g.

$$g(x) = x - f(x) \quad \text{or} \quad g(x) = x + 3f(x).$$

Then $g(p) = p$ whenever $f(p) = 0$.

- Conversely, if g has a fixed point p ($g(p) = p$), then

$$f(x) = x - g(x)$$

satisfies $f(p) = 0$.

Remarks. Although our goal is often to solve $f(x) = 0$, the fixed-point form can be easier to analyze, and suitable choices of g lead to powerful root-finding schemes.

Example. Determine the fixed points of $g(x) = x^2 - x$.

Solution. A fixed point p satisfies

$$p = g(p) = p^2 - p \implies 0 = p^2 - 2p = p(p - 2).$$

Hence the fixed points are

$$p = 0 \quad \text{and} \quad p = 2,$$

i.e., the intersections of $y = g(x)$ with $y = x$.

Theorem 2 (Fixed Point Existence and Uniqueness). (i) If $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.

(ii) If, in addition, $g'(x)$ exists on (a, b) and there is a constant $k < 1$ such that

$$|g'(x)| \leq k \quad \text{for all } x \in (a, b),$$

then there is exactly one fixed point in $[a, b]$.

Proof. (i) If $g(a) = a$ or $g(b) = b$, then g has a fixed point at an endpoint. Otherwise $g(a) > a$ and $g(b) < b$. Define $h(x) = g(x) - x$. Then h is continuous on $[a, b]$, with

$$h(a) = g(a) - a > 0, \quad h(b) = g(b) - b < 0.$$

By the Intermediate Value Theorem there exists $p \in (a, b)$ such that $h(p) = 0$. Hence $0 = h(p) = g(p) - p$, i.e., $g(p) = p$.

(ii) Assume in addition that $|g'(x)| \leq k < 1$ on (a, b) and suppose p and q are both fixed points in $[a, b]$. If $p \neq q$, then by the Mean Value Theorem there exists ξ between p and q (hence $\xi \in (a, b)$) such that

$$\frac{g(p) - g(q)}{p - q} = g'(\xi).$$

Therefore

$$|p - q| = |g(p) - g(q)| = |g'(\xi)| |p - q| \leq k |p - q| < |p - q|,$$

a contradiction. Thus $p = q$, and the fixed point in $[a, b]$ is unique. $\square$

Highlight: IVT and MVT (as they appear many times). Highlight how derivative conditions are very useful for ensuring uniqueness as we've also seen when discussing root finding.

Example 2 Show that $g(x) = \frac{x^2 - 1}{3}$ has a unique fixed point on the interval $[-1, 1]$.

Solution. The extreme values of g on $[-1, 1]$ occur at endpoints or where $g'(x) = 0$. Since $g'(x) = \frac{2x}{3}$, g is continuous on $[-1, 1]$ and differentiable on $(-1, 1)$, with critical point $x = 0$ and endpoints $x = \pm 1$. Compute

$$g(-1) = 0, \quad g(1) = 0, \quad g(0) = -\frac{1}{3}.$$

Hence $g([-1, 1]) \subset [-\frac{1}{3}, 0] \subset [-1, 1]$.

Moreover,

$$|g'(x)| = \left| \frac{2x}{3} \right| \leq \frac{2}{3} < 1 \quad \text{for all } x \in (-1, 1).$$

Thus g satisfies all hypotheses of Theorem 2: it maps $[-1, 1]$ into itself and $|g'| \leq k < 1$ on $(-1, 1)$. Therefore g has *exactly one* fixed point in $[-1, 1]$.

Fixed Point Approximation

To approximate a fixed point of a function g , choose an initial guess p_0 and generate the sequence $\{p_n\}_{n=0}^\infty$ by

$$p_n = g(p_{n-1}), \quad n \geq 1.$$

If the sequence converges to p and g is continuous, then

$$p = \lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} g(p_{n-1}) = g\left(\lim_{n \rightarrow \infty} p_{n-1}\right) = g(p),$$

so p solves $x = g(x)$. This procedure is called *fixed-point* (or *functional*) iteration.

[H] **Fixed-Point (Functional) Iteration for $p = g(p)$**

Require: function g , initial guess p_0 , tolerance TOL, max iterations $N_{\max}$

Ensure: approximate fixed point p or failure

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1:  $i \leftarrow 1$ 
2: while  $i \leq N_{\max}$  do
3:    $p \leftarrow g(p_0)$  ▷ compute  $p_i$ 
4:   if  $|p - p_0| < \text{TOL}$  then
5:     return  $p$  ▷ success
6:   end if
7:    $p_0 \leftarrow p$  ▷ update iterate
8:    $i \leftarrow i + 1$ 
9: end while
10: return failure: no convergence within  $N_{\max}$  iterations
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Theorem 3 (Convergence of Fixed-Point Iteration). Let $g \in C[a, b]$ and suppose $g(x) \in [a, b]$ for all $x \in [a, b]$. Assume, in addition, that g' exists on (a, b) and there is a constant $0 < k < 1$ with

$$|g'(x)| \leq k, \quad \text{for all } x \in (a, b).$$

Then, for any $p_0 \in [a, b]$, the sequence defined by

$$p_n = g(p_{n-1}), \quad n \geq 1,$$

converges to the unique fixed point $p \in [a, b]$.

Proof. By Theorem 2, there exists a unique $p \in [a, b]$ with $g(p) = p$. Since g maps $[a, b]$ into itself, the sequence $\{p_n\}_{n=0}^\infty$ is well-defined and $p_n \in [a, b]$ for all n . Using $|g'(x)| \leq k$ and the Mean Value Theorem, for each n there exists $\xi_n \in (a, b)$ such that

$$|p_n - p| = |g(p_{n-1}) - g(p)| = |g'(\xi_n)| |p_{n-1} - p| \leq k |p_{n-1} - p|.$$

Applying this inductively yields

$$|p_n - p| \leq k |p_{n-1} - p| \leq k^2 |p_{n-2} - p| \leq \cdots \leq k^n |p_0 - p|. \quad (2.4)$$

Since $0 < k < 1$, we have $\lim_{n \rightarrow \infty} k^n = 0$, hence

$$\lim_{n \rightarrow \infty} |p_n - p| \leq \lim_{n \rightarrow \infty} k^n |p_0 - p| = 0.$$

Therefore $p_n \rightarrow p$, as claimed. $\square$

If g satisfies the hypotheses of Theorem 2 and the convergence theorem above (i.e., $g : [a, b] \rightarrow [a, b]$, $g \in C[a, b]$, and $|g'| \leq k < 1$ on (a, b)), then the errors of fixed-point iteration $p_n = g(p_{n-1})$ satisfy

$$|p_n - p| \leq k^n \max\{p_0 - a, b - p_0\}, \quad (2.5)$$

and, for all $n \geq 1$,

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|. \quad (2.6)$$

Take-away. How can we choose a fixed-point formulation that reliably and rapidly converges to the solution of a given root-finding problem?

Answer. Rewrite the equation $f(x) = 0$ as $x = g(x)$ so that g maps $[a, b]$ into itself and satisfies the hypotheses of the fixed-point theorem (continuous, $|g'(x)| \leq k < 1$ near the solution). Aim to make $|g'(p)|$ as small as possible to accelerate convergence.

In the next sections we examine constructions of such g in detail.

Section 2.3/2.4 Newton's Method and Its Extensions

Assume $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to a root p with $f'(p_0) \neq 0$ and $|p - p_0|$ small. The first-order Taylor expansion of f about p_0 , evaluated at $x = p$, gives

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)),$$

where $\xi(p)$ lies between p and p_0 . Since $f(p) = 0$,

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)).$$

If $|p - p_0|$ is small, the quadratic term is much smaller, so

$$0 \approx f(p_0) + (p - p_0)f'(p_0).$$

Solving for p yields the Newton update

$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} \equiv p_1.$$

Iteration. Starting from p_0 , define $\{p_n\}_{n=0}^\infty$ by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1. \quad (2.7)$$

[H] **Newton's Method**

Require: function f , derivative f' , initial guess p_0 , tolerance TOL, max iterations $N_{\max}$

Ensure: approximate root p or failure

1: $i \leftarrow 1$

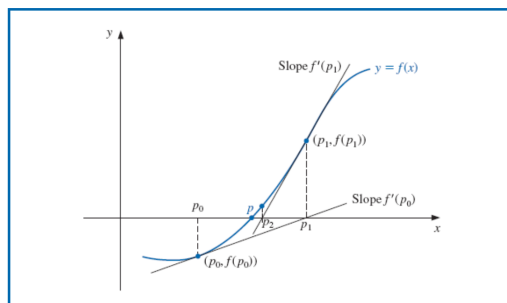
2: **while** $i \leq N_{\max}$ **do**

3: $p \leftarrow p_0 - \frac{f(p_0)}{f'(p_0)}$

$\triangleright$ Newton update

Figure 2.7 illustrates how the approximations are obtained using successive tangents. (Also see Exercise 31.) Starting with the initial approximation p_0 , the approximation p_1 is the x -intercept of the tangent line to the graph of f at $(p_0, f(p_0))$. The approximation p_2 is the x -intercept of the tangent line to the graph of f at $(p_1, f(p_1))$ and so on. Algorithm 2.3 implements this procedure.

Figure 2.7



► Details

```

4:  if  $|p - p_0| < \text{TOL}$  or  $|f(p)| < \text{TOL}$  then
5:      return  $p$ 
6:  end if
7:   $p_0 \leftarrow p$ 
8:   $i \leftarrow i + 1$ 
9: end while
10: return failure: no convergence within  $N_{\max}$  iterations

```

► success

Theorem 4 (Local convergence of Newton's method). Let $f \in C^2[a, b]$. If $p \in (a, b)$ satisfies $f(p) = 0$ and $f'(p) \neq 0$, then there exists $\delta > 0$ such that Newton's method generates a sequence $\{p_n\}_{n=1}^{\infty}$ converging to p for any initial approximation $p_0 \in [p - \delta, p + \delta]$.

Proof. View Newton's method as the functional iteration $p_n = g(p_{n-1})$ with

$$g(x) = x - \frac{f(x)}{f'(x)}, \quad n \geq 1.$$

Choose $k \in (0, 1)$. Since f' is continuous and $f'(p) \neq 0$, there exists $\delta_1 > 0$ such that $f'(x) \neq 0$ for $x \in [p - \delta_1, p + \delta_1] \subset (a, b)$. Hence g is defined and continuous on $[p - \delta_1, p + \delta_1]$. Moreover,

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

so, using $f(p) = 0$,

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

Because g' is continuous and $0 < k < 1$, there exists δ with $0 < \delta < \delta_1$ such that

$$|g'(x)| \leq k, \quad \text{for all } x \in [p - \delta, p + \delta].$$

It remains to show that g maps $[p - \delta, p + \delta]$ into itself. For $x \in [p - \delta, p + \delta]$, the Mean Value Theorem gives

a ξ between x and p with

$$|g(x) - p| = |g(x) - g(p)| = |g'(\xi)| |x - p| \leq k |x - p| < |x - p|.$$

Since $x \in [p - \delta, p + \delta]$, we have $|x - p| < \delta$, hence $|g(x) - p| < \delta$, so $g([p - \delta, p + \delta]) \subset [p - \delta, p + \delta]$.

All hypotheses of the fixed-point contraction theorem now hold; therefore, the iteration

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \quad n \geq 1,$$

converges to p for any $p_0 \in [p - \delta, p + \delta]$. □

Newton's Method Assume $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to a root p with $f'(p_0) \neq 0$ and $|p - p_0|$ small. The first-order Taylor expansion of f about p_0 and solving for p , gives the Newton update.

Last time we saw that if the initial guess p_0 is sufficiently close to p then the method will converge. Today we will answer – how close does it need to be.

The takeaway will be that similar to how first derivatives controlled behavior of bisection methods, now for the second-order Newton's method the first and second-order derivatives will be used to determine "how close".

Theorem 5 (Newton step and local error). Assume $f \in C^2[a, b]$ and $p \in [a, b]$ is a simple root: $f(p) = 0$, $f'(p) \neq 0$. For p_0 sufficiently close to p , define the Newton iterates

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}.$$

Then for each n there exists ξ_n between p_n and p such that

$$p - p_{n+1} = -\frac{f''(\xi_n)}{2f'(p_n)}(p - p_n)^2.$$

This follows directly from the Taylor series expansion.

Theorem 6 (Quadratic convergence of Newton's method). Assume $f \in C^2[a, b]$ and let $p \in (a, b)$ satisfy $f(p) = 0$ and $f'(p) \neq 0$. If p_0 is sufficiently close to p , the Newton iterates

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}, \quad n \geq 0,$$

are well defined, satisfy $p_n \rightarrow p$, and

$$\lim_{n \rightarrow \infty} \frac{p - p_{n+1}}{(p - p_n)^2} = -\frac{f''(p)}{2f'(p)}.$$

In particular, the order of convergence is $p = 2$.

Proof. Since $f'(p) \neq 0$ and f' is continuous, choose $\varepsilon > 0$ with $I = [p - \varepsilon, p + \varepsilon] \subset (a, b)$ and $\min_{x \in I} |f'(x)| > 0$. Define

$$M := \frac{\max_{x \in I} |f''(x)|}{2 \min_{x \in I} |f'(x)|}.$$

By Taylor's theorem with Lagrange remainder, for $p_n \in I$ there exists ξ_n between p_n and p such that

$$p - p_{n+1} = -\frac{f''(\xi_n)}{2f'(p_n)}(p - p_n)^2.$$

Taking absolute values and using the definition of M yields the key estimate

$$|p - p_{n+1}| \leq M |p - p_n|^2. \quad (1)$$

Staying in I and convergence. Pick $p_0 \in I$ so close to p that $|p - p_0| \leq \varepsilon$ and $M|p - p_0| < 1$. Then (1) gives $M|p - p_1| \leq (M|p - p_0|)^2 < M|p - p_0| \leq 1$, hence $|p - p_1| \leq \varepsilon$; inductively, $p_n \in I$ for all n , and

$$M|p - p_n| \leq (M|p - p_0|)^{2^n}.$$

Thus $|p - p_n| \rightarrow 0$, i.e. $p_n \rightarrow p$.

Asymptotic constant. Because $p_n \rightarrow p$ and ξ_n lies between p_n and p , we have $\xi_n \rightarrow p$. Continuity of f' and f'' then gives $f'(\xi_n) \rightarrow f'(p)$ and $f''(\xi_n) \rightarrow f''(p)$. Divide the exact error identity by $(p - p_n)^2$ to obtain

$$\frac{p - p_{n+1}}{(p - p_n)^2} = -\frac{f''(\xi_n)}{2f'(\xi_n)} \rightarrow -\frac{f''(p)}{2f'(p)}.$$

This proves quadratic convergence with the stated limit. $\square$

Lemma 1 (Practical convergence test for Newton). Let $f \in C^2[a, b]$ and $p \in (a, b)$ with $f(p) = 0$, $f'(p) \neq 0$. Choose $I = [p - \varepsilon, p + \varepsilon] \subset (a, b)$ so that

$$m := \min_{x \in I} |f'(x)| > 0, \quad M_2 := \max_{x \in I} |f''(x)| < \infty,$$

and set

$$M := \frac{M_2}{2m}.$$

If the initial guess $p_0 \in I$ satisfies

$$|p - p_0| < \frac{1}{M},$$

then the Newton iterates $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$ stay in I and converge to p , with

$$|p - p_{n+1}| \leq M |p - p_n|^2.$$

Interpretation: larger M (large curvature or small slope) requires a closer initial guess.

The Secant Method

Newton's method is powerful but requires evaluating f' at each step. To avoid derivative evaluations, approximate f' by a secant slope. By definition,

$$f'(p_{n-1}) = \lim_{x \rightarrow p_{n-1}} \frac{f(x) - f(p_{n-1})}{x - p_{n-1}}.$$

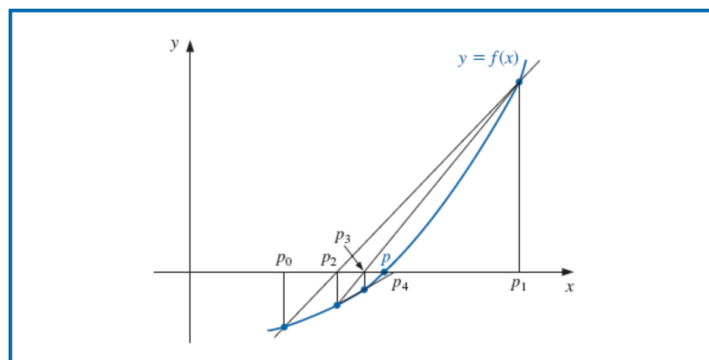
If p_{n-2} is close to p_{n-1} , then

$$f'(p_{n-1}) \approx \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}.$$

Definition. Substituting this in Newton's update yields the *secant* iteration

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}, \quad n \geq 2. \quad (2)$$

Figure 2.9



▼ Details

A curve is graphed on an x y coordinate plane. The x axis has tick marks labeled from the left to the right as follows: p_0 , p_2 , p_3 , p_4 , and p_1 . 5 points are plotted, in which a point each is plotted directly below p_0 , p_2 , and p_3 , respectively. A point p is marked on the x axis between p_3 and p_4 , close to p_4 . The fifth point is marked directly above p_1 . The curve labeled $y = f(x)$ rises as a concave up increasing curve from the fourth quadrant to the first quadrant through the 5 points. Four vertical lines are drawn from p_0 , p_2 , p_3 , and p_1 on the x axis to the curve. Two upward sloping lines are drawn from the plotted points below $x = p_0$ and p_2 in the fourth quadrant to the plotted point above $x = p_1$ in the first quadrant through p_2 and p_3 on the x axis, respectively. A small line slopes upward from the plotted point below $x = p_2$ to the point p_4 on the x axis.

Geometric interpretation. Starting from two initial approximations p_0 and p_1 :

- p_2 is the x -intercept of the line through $(p_0, f(p_0))$ and $(p_1, f(p_1))$.
- p_3 is the x -intercept of the line through $(p_1, f(p_1))$ and $(p_2, f(p_2))$, etc.

The Method of False Position

Idea. Like the Secant method, use the secant line to estimate the root, but *always keep the root bracketed*. Start with a, b such that $f(a)f(b) < 0$. At each step form the secant through $(a, f(a))$ and $(b, f(b))$ and take its x -intercept as the next approximation. Replace the endpoint that has the *same* sign as the new value so that the sign change (and hence the bracket) is preserved.

Secant intercept. Given current bracket $[a, b]$ with $f(a)f(b) < 0$, define

$$p = b - \frac{f(b)(b-a)}{f(b)-f(a)} = a - \frac{f(a)(a-b)}{f(a)-f(b)}.$$

Bracketing update.

If $f(a)f(p) < 0$ then $b \leftarrow p$; else $a \leftarrow p$.

This guarantees $f(a)f(b) < 0$ after every iteration.

[H] **False Position**

Require: function f , initial a, b with $f(a)f(b) < 0$, tolerance TOL, max its $N_{\max}$

Ensure: approximate root p (bracket maintained) or failure

```

1: for  $n = 1$  to  $N_{\max}$  do
2:    $p \leftarrow b - \frac{f(b)(b-a)}{f(b)-f(a)}$  ▷  $x$ -intercept of secant
3:   if  $|f(p)| < \text{TOL}$  or  $\max(|p-a|, |b-p|) < \text{TOL}$  then
4:     return  $p$ 
5:   end if
6:   if  $f(a)f(p) < 0$  then
7:      $b \leftarrow p$ 
8:   else
9:      $a \leftarrow p$ 
10:  end if
11: end for
12: return failure: no convergence within  $N_{\max}$  iterations

```

Remarks.

- Each step needs only one new function evaluation (after the first two).
- Bracketing is preserved, so a root remains in $[a, b]$; however, unlike bisection, the interval length need not halve each step.

Theorem 7 (**Steffensen's method is quadratic (limit p)**). Let $g \in C^2$ in a neighborhood of a fixed point p with $g(p) = p$ and $\lambda := g'(p)$ satisfying $|\lambda| < 1$ and $\lambda \neq 1$. Define Steffensen's iteration

$$x_{n+1} = x_n - \frac{(g(x_n) - x_n)^2}{g(g(x_n)) - 2g(x_n) + x_n}.$$

For x_0 sufficiently close to p , the iterates $x_n \rightarrow p$ and

$$x_{n+1} - p = \frac{g''(p)}{2(1-\lambda)} (x_n - p)^2 + o((x_n - p)^2),$$

i.e. the convergence is quadratic.

Sketch. Let $e_n := x_n - p$. Taylor at p :

$$g(p+e) = p + \lambda e + ce^2 + o(e^2), \quad c := \frac{1}{2}g''(p).$$

Then $g(x_n) - x_n = (\lambda - 1)e_n + ce_n^2 + o(e_n^2)$ and $g(g(x_n)) = p + \lambda^2 e_n + c(\lambda + \lambda^2)e_n^2 + o(e_n^2)$. Hence

$$D := g(g(x_n)) - 2g(x_n) + x_n = (1 - \lambda)^2 e_n + O(e_n^2), \quad N := (g(x_n) - x_n)^2 = (1 - \lambda)^2 e_n^2 + O(e_n^3).$$

So

$$\frac{N}{D} = e_n - \frac{2c}{1-\lambda} e_n^2 + o(e_n^2),$$

and

$$e_{n+1} = e_n - \frac{N}{D} = \frac{2c}{1-\lambda} e_n^2 + o(e_n^2) = \frac{g''(p)}{2(1-\lambda)} e_n^2 + o(e_n^2).$$

□

Theorem 8 (Steffensen's method is quadratic). Let $g \in C^2$ in a neighborhood of a fixed point α with $g(\alpha) = \alpha$ and $\lambda := g'(\alpha)$ satisfying $|\lambda| < 1$ and $\lambda \neq 1$. Define Steffensen's iteration

$$x_{n+1} = x_n - \frac{(g(x_n) - x_n)^2}{g(g(x_n)) - 2g(x_n) + x_n}.$$

Then for x_0 sufficiently close to α , the iterates $x_n \rightarrow \alpha$ and

$$x_{n+1} - \alpha = \frac{g''(\alpha)}{2(1-\lambda)} (x_n - \alpha)^2 + o((x_n - \alpha)^2),$$

i.e. the convergence is quadratic.

Sketch. Write $e_n := x_n - \alpha$. Taylor-expand g at α :

$$g(\alpha + e) = \alpha + \lambda e + ce^2 + o(e^2), \quad c := \frac{1}{2}g''(\alpha).$$

Hence

$$g(x_n) - x_n = (\lambda - 1)e_n + ce_n^2 + o(e_n^2).$$

Apply g again with $\delta := \lambda e_n + ce_n^2 + o(e_n^2)$:

$$g(g(x_n)) = \alpha + \lambda\delta + c\delta^2 + o(e_n^2) = \alpha + \lambda^2 e_n + c(\lambda + \lambda^2)e_n^2 + o(e_n^2).$$

Therefore the denominator

$$D := g(g(x_n)) - 2g(x_n) + x_n = (1 - \lambda)^2 e_n + O(e_n^2),$$

and the numerator

$$N := (g(x_n) - x_n)^2 = (1 - \lambda)^2 e_n^2 + O(e_n^3).$$

Thus

$$\frac{N}{D} = \frac{(1 - \lambda)^2 e_n^2 + O(e_n^3)}{(1 - \lambda)^2 e_n + O(e_n^2)} = e_n - \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2).$$

Finally,

$$e_{n+1} = x_{n+1} - \alpha = e_n - \frac{N}{D} = \frac{2c}{1 - \lambda} e_n^2 + o(e_n^2) = \frac{g''(\alpha)}{2(1 - \lambda)} e_n^2 + o(e_n^2),$$

which proves quadratic convergence. □

Section 2.4/2.5/2.6 (Text only)

Section 3.1 Interpolation

Vandermonde formulation. Suppose we seek a polynomial

$$p(x) = a_0 + a_1 x + \cdots + a_m x^m.$$

Since (3.1.1) imposes $n + 1$ conditions on $p(x)$, it is reasonable to first consider the case $m = n$. Then we want to find $a_0, a_1, \dots, a_n$ such that

$$\begin{aligned} a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n &= y_0, \\ &\vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \cdots + a_n x_n^n &= y_n. \end{aligned}$$

This is a system of $n + 1$ linear equations in $n + 1$ unknowns, completely equivalent to solving the polynomial interpolation problem. In vector–matrix notation, the system is

$$X a = y$$

with

$$\begin{aligned} X_{i,j} &= [x_i^j], \quad i, j = 0, 1, \dots, n, \\ a &= [a_0 \ a_1 \ \cdots \ a_n]^\top, \quad y = [f(x_0) \ \cdots \ f(x_n)]^\top. \end{aligned}$$

The matrix X is called a *Vandermonde matrix*.

Section 3.2/3.3/3.4 (Text only)

Section 3.5 Cubic Spline

Example (Cubic Hermite interpolation). The most widely used form of Hermite interpolation is the cubic Hermite polynomial, which solves

$$p(a) = f(a), \quad p'(a) = f'(a), \quad p(b) = f(b), \quad p'(b) = f'(b). \quad (3)$$

The divided–difference form becomes

$$H_2(x) = f(a) + (x - a)f'(a) + (x - a)^2 f[a, a, b] + (x - a)^2 (x - b) f[a, a, b, b], \quad (4)$$

in which

$$f[a, a, b] = \frac{f[a, b] - f'(a)}{b - a}, \quad f[a, a, b, b] = \frac{f'(b) - 2f[a, b] + f'(a)}{(b - a)^2}.$$

The error formula is

$$\begin{aligned} f(x) - H_2(x) &= (x - a)^2 (x - b)^2 f[a, a, b, b, x] \\ &= \frac{(x - a)^2 (x - b)^2}{24} f^{(4)}(\xi_x), \end{aligned} \quad (5)$$

and hence

$$\max_{a \leq x \leq b} |f(x) - H_2(x)| \leq \frac{(b - a)^4}{384} \max_{a \leq t \leq b} |f^{(4)}(t)|. \quad (6)$$

Notation. For a piecewise polynomial function $p(x)$, there is an associated grid

$$-\infty < x_0 < x_1 < \cdots < x_n < \infty. \quad (7)$$

The points x_j are sometimes called *knots*, *breakpoints*, or *nodes*. The function $p(x)$ is a polynomial on each of the subintervals

$$(-\infty, x_0], [x_0, x_1], \dots, [x_{n-1}, x_n], [x_n, \infty), \quad (8)$$

although often the end intervals $(-\infty, x_0]$ and $[x_n, \infty)$ are not included. We say $p(x)$ is a *piecewise polynomial of order r* if the degree of $p(x)$ is at most r on each of the subintervals. Unless specified otherwise, no continuity across the knots is required.

Spline functions. As before, consider a grid

$$a = x_0 < x_1 < \cdots < x_n = b.$$

We say $s(x)$ is a *spline function of order $m \geq 1$* if it satisfies:

(P1) On each subinterval $[x_{i-1}, x_i]$, $s(x)$ is a polynomial of degree $< m$.

(P2) For $0 \leq r \leq m-2$, the derivative $s^{(r)}(x)$ is continuous on $[a, b]$ (i.e., $s \in C^{m-2}[a, b]$).

The derivative of a spline of order m is a spline of order $m-1$ (and similarly antidifferentiation raises the order by one). Moreover, if the continuity in (P2) is strengthened to include $s^{(m-1)}(x)$ (i.e., $s \in C^{m-1}[a, b]$), then $s(x)$ must in fact be a single polynomial on $[a, b]$ of degree $\leq m-1$.

Degrees of Freedom. For our interpolation problem, we seek a *cubic spline* $s(x)$ such that

$$s(x_i) = y_i, \quad i = 0, 1, \dots, n. \quad (9)$$

On each subinterval $[x_{i-1}, x_i]$ we write

$$s(x) = a_i + b_i x + c_i x^2 + d_i x^3, \quad i = 1, \dots, n. \quad (10)$$

There are $4n$ unknown coefficients $\{a_i, b_i, c_i, d_i\}$. The constraints are the interpolation conditions (3.7.11) together with the spline continuity conditions (from C^2 -continuity):

$$s^{(j)}(x_i + 0) = s^{(j)}(x_i - 0), \quad i = 1, \dots, n-1, \quad j = 0, 1, 2. \quad (11)$$

Counting equations gives

$$(n+1) + 3(n-1) = 4n-2,$$

compared with $4n$ unknowns. Hence two additional conditions (boundary conditions) are required to determine a unique cubic interpolating spline $s(x)$.

Oscillatory Behaviors. Let $f \in C^4[a, b]$ and let $x_0 = a < x_1 < \cdots < x_n = b$ be a partition. Let s_c be the *complete (clamped) cubic spline* interpolant of f satisfying

$$s_c(x_i) = f(x_i) \quad (i = 0, \dots, n), \quad s'_c(a) = f'(a), \quad s'_c(b) = f'(b).$$

Consider the admissible class

$$\mathcal{A} := \{g \in C^2[a, b] : g(x_i) = f(x_i) \quad (i = 0, \dots, n), \quad g'(a) = f'(a), \quad g'(b) = f'(b)\}.$$

Then

$$\int_a^b |s''_c(x)|^2 dx \leq \int_a^b |g''(x)|^2 dx, \quad \forall g \in \mathcal{A}, \quad (12)$$

with equality iff $g \equiv s_c$.

Proof. Let $g \in \mathcal{A}$ and set $k := s_c - g$. Then

$$k(x_i) = 0 \quad (i = 0, \dots, n), \quad k'(a) = k'(b) = 0.$$

Expanding the square,

$$\int_a^b |g''|^2 = \int_a^b |s_c'' - k''|^2 = \int_a^b |s_c''|^2 - 2 \int_a^b s_c'' k'' + \int_a^b |k''|^2.$$

Using integration by parts one can find

$$\int_a^b s_c''(x) k''(x) dx = 0. \quad (13)$$

Indeed, integrate by parts on each subinterval $[x_{i-1}, x_i]$ and sum:

$$\int_{x_{i-1}}^{x_i} s_c'' k'' dx = \left[s_c'' k' - s_c''' k \right]_{x_{i-1}}^{x_i} - \int_{x_{i-1}}^{x_i} s_c^{(4)} k dx.$$

Since s_c is cubic on each $[x_{i-1}, x_i]$, $s_c^{(4)} \equiv 0$ there. Summing over $i = 1, \dots, n$ gives telescoping boundary terms. Using that s_c'' is continuous at each knot, $k(x_i) = 0$ for all i , and $k'(a) = k'(b) = 0$, all boundary contributions vanish; hence (??).

Therefore,

$$\int_a^b |g''|^2 = \int_a^b |s_c''|^2 + \int_a^b |k''|^2 \geq \int_a^b |s_c''|^2. \quad (14)$$

Equality holds iff $k'' \equiv 0$, i.e. k is linear. The interpolation and clamped boundary conditions then force $k \equiv 0$, so $g \equiv s_c$. $\square$

Exploring *not-a-knot* condition.

Theorem 9 (Error of the complete (clamped) cubic spline). Let $f \in C^4[a, b]$. For each n , let a partition

$$\tau_n : \quad a = x_0^{(n)} < x_1^{(n)} < \dots < x_n^{(n)} = b,$$

and define the mesh width

$$|\tau_n| = \max_{1 \leq i \leq n} (x_i^{(n)} - x_{i-1}^{(n)}).$$

Let $s_{c,n}$ be the complete (clamped) cubic spline interpolant of f on τ_n , i.e.

$$s_{c,n}(x_i^{(n)}) = f(x_i^{(n)}) \quad (i = 0, 1, \dots, n), \quad s'_{c,n}(a) = f'(a), \quad s'_{c,n}(b) = f'(b).$$

Then there exist absolute constants c_j such that for $j = 0, 1, 2$,

$$\max_{a \leq x \leq b} |f^{(j)}(x) - s_{c,n}^{(j)}(x)| \leq c_j |\tau_n|^{4-j} \max_{a \leq x \leq b} |f^{(4)}(x)|. \quad (15)$$

Acceptable constants are

$$c_0 = \frac{5}{384}, \quad c_1 = \frac{1}{24}, \quad c_2 = \frac{3}{8}.$$

Section 4.1 Forward/backward differences

Numerical Differentiation

One of the main approaches to deriving a numerical approximation to $f'(x)$ is to use the derivative of a polynomial $p_n(x)$ that interpolates $f(x)$ at a given set of node points. Let $x_0, x_1, \dots, x_n$ be given, and let $p_n(x)$ interpolate $f(x)$ at these nodes. Usually $\{x_i\}$ are evenly spaced.

Key Definitions

$$f'(x) \approx p'_n(x).$$

$$p_n(x) = \sum_{j=0}^n f(x_j) l_j(x),$$

$$l_j(x) = \frac{\Psi_n(x)}{(x - x_j) \Psi'_n(x_j)},$$

$$l_j(x) = \frac{(x - x_0) \cdots (x - x_{j-1})(x - x_{j+1}) \cdots (x - x_n)}{(x_j - x_0) \cdots (x_j - x_{j-1})(x_j - x_{j+1}) \cdots (x_j - x_n)},$$

$$\Psi_n(x) = (x - x_0) \cdots (x - x_n),$$

$$f(x) - p_n(x) = \Psi_n(x) f[x_0, \dots, x_n, x].$$

Derivative Approximation using Interpolation

Thus

$$f'(x) \approx p'_n(x) = \sum_{j=0}^n f(x_j) l'_j(x) \equiv D_h f(x).$$

$$f'(x) - p'_n(x) = \Psi'_n(x) f[x_0, \dots, x_n, x] + \Psi_n(x) f[x_0, \dots, x_n, x, x].$$

From the definitions above

$$f'(x) - p'_n(x) = \Psi'_n(x) \frac{f^{(n+1)}(\xi_1)}{(n+1)!} + \Psi_n(x) \frac{f^{(n+2)}(\xi_2)}{(n+2)!},$$

with $\xi_1, \xi_2 \in \mathcal{X}\{x_0, \dots, x_n, x\}$. Higher order differentiation formulas and their error can be obtained by further differentiation.

The most common application of the preceding is to evenly spaced nodes $\{x_i\}$. Thus let

$$x_i = x_0 + ih, \quad i \geq 0,$$

with $h > 0$. In this case, it is straightforward to show that

$$\Psi_n(x) = \mathcal{O}(h^{n+1}), \quad \Psi'_n(x) = \mathcal{O}(h^n).$$

Thus

$$f'(x) - p'_n(x) = \begin{cases} \mathcal{O}(h^n) & \Psi'_n(x) \neq 0, \\ \mathcal{O}(h^{n+1}) & \Psi'_n(x) = 0. \end{cases}$$

Example 1. Let $n = 1$, so that $p_n(x)$ is just the linear interpolate of $(x_0, f(x_0))$ and $(x_1, f(x_1))$. Then (5.7.3) yields

$$f'(x_0) \approx p'_n(x_0) \equiv \frac{1}{h} [f(x_0 + h) - f(x_0)].$$

$$f'(x_0) - p'_n(x_0) = \frac{h}{2} f''(\xi_1), \quad x_0 \leq \xi_1 \leq x_1,$$

since $\Psi(x_0) = 0$.

Example 2. To improve on this with linear interpolation, choose $x = m \equiv (x_0 + x_1)/2$. Then

$$f'(m) \approx \frac{1}{h} [f(x_1) - f(x_0)].$$

We usually rewrite this by letting $\delta = h/2$, to obtain

$$f'(m) \approx D_\delta f(m) = \frac{1}{2\delta} [f(m + \delta) - f(m - \delta)].$$

For the error,

$$f'(m) - D_\delta f(m) = -\frac{\delta^2}{6} f^{(3)}(\xi_2), \quad m - \delta \leq \xi_2 \leq m + \delta.$$

In general, to obtain the higher-order case, we want to choose the nodes $\{x_i\}$ to have $\Psi'_n(x) = 0$. This will be true if n is odd and the nodes are placed symmetrically about x .

To obtain higher-order formulas in which the nodes all lie on one side of x , use higher values of n . For example, with $x = x_0$ and $n = 2$,

$$f'(x_0) \approx D_h f(x_0) \equiv \frac{1}{2h} [-3f(x_0) + 4f(x_1) - f(x_2)],$$

$$f'(x_0) - D_h f(x_0) = \frac{h^2}{3} f^{(3)}(\xi_1), \quad x_0 \leq \xi_1 \leq x_2.$$

Section 4.1 Finite-difference formulas

Aitken Extrapolation for Sequences

Reconsider

$$\lim_{n \rightarrow \infty} \frac{x - x_{n+1}}{x - x_n} = g'(x).$$

for a convergent iteration

$$x_{n+1} = g(x_n), \quad n \geq 0.$$

Considering only linear convergent sequences.

Thus we will assume

$$0 < |g'(x)| < 1.$$

We examine estimating the error in the iterates and give a way to accelerate the convergence of $\{x_n\}$. We begin by considering the ratios

$$\lambda_n = \frac{x_n - x_{n-1}}{x_{n-1} - x_{n-2}}, \quad n \geq 2.$$

Claim:

$$\lim_{n \rightarrow \infty} \lambda_n = g'(x).$$

To see this, write

$$\lambda_n = \frac{(x - x_{n-1}) - (x - x_n)}{(x - x_{n-2}) - (x - x_{n-1})}.$$

Then

$$\lambda_n = \frac{(x - x_{n-1}) - g'(\xi_{n-1})(x - x_{n-1})}{(x - x_{n-2})/[g'(\xi_{n-2})] - (x - x_{n-1})} = \frac{1 - g'(\xi_{n-1})}{1/[g'(\xi_{n-2})] - 1}.$$

Hence

$$\lim_{n \rightarrow \infty} \lambda_n = \frac{1 - g'(x)}{1/[g'(x)] - 1} = g'(x).$$

The quantity λ_n is computable, and when it converges empirically to a value λ , we assume $\lambda = g'(x)$. We use $\lambda_n \approx g'(x)$ to estimate the error in the iterates x_n . Assume

$$x - x_n \doteq \lambda_n(x - x_{n-1}).$$

Then

$$x - x_n = (x - x_{n-1}) + (x_{n-1} - x_n) \doteq \frac{1}{\lambda_n}(x - x_n) + (x_{n-1} - x_n),$$

so that

$$x - x_n \doteq \frac{\lambda_n}{1 - \lambda_n} (x_n - x_{n-1})$$

The Trapezoidal Rule

The simple trapezoidal rule is based on approximating $f(x)$ by the straight line joining $(a, f(a))$ and $(b, f(b))$. By integrating the formula for this straight line, we obtain the approximation

$$I_1(f) = \left(\frac{b-a}{2} \right) [f(a) + f(b)]$$

To obtain an error formula, we use the interpolation error formula

$$f(x) - \frac{(b-x)f(a) + (x-a)f(b)}{b-a} = (x-a)(x-b)f[a, b, x]$$

We also assume for all work with the error for the trapezoidal rule in this section that $f(x)$ is twice continuously differentiable on $[a, b]$. Then

$$\begin{aligned} E_1(f) &= \int_a^b f(x) dx - \frac{(b-a)}{2} [f(a) + f(b)] \\ &= \int_a^b (x-a)(x-b)f[a, b, x] dx \end{aligned}$$

Using the integral mean value theorem.

$$E_1(f) = f[a, b, \xi] \int_a^b (x-a)(x-b) dx, \quad \text{for some } a \leq \xi \leq b$$

$$= \left[\frac{1}{2} f''(\eta) \right] \left[-\frac{1}{6} (b-a)^3 \right], \quad \text{for some } \eta \in [a, b].$$

Simpson's rule

To improve upon the simple trapezoidal rule, we use a quadratic interpolating polynomial $p_2(f)$ to approximate $f(x)$ on $[a, b]$. Let $c = (a+b)/2$, and define

$$I_2(f) = \int_a^b \left[\frac{(x-c)(x-b)}{(a-c)(a-b)} f(a) + \frac{(x-a)(x-b)}{(c-a)(c-b)} f(c) + \frac{(x-a)(x-c)}{(b-a)(b-c)} f(b) \right] dx.$$

Carrying out the integration, we obtain

$$I_2(f) = \frac{h}{3} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right], \quad h = \frac{b-a}{2}.$$

This is called *Simpson's rule*. An illustration is given in Figure 5.2, with the shaded region denoting the area under the graph of $y = p_2(x)$.

For the error, we begin with the interpolation error formula (3.2.11) to obtain

$$E_2(f) = I(f) - I_2(f)$$

We cannot apply the integral mean value theorem since the polynomial in the integrand changes sign at $x = c = (a+b)/2$. We will assume $f(x)$ is four times continuously differentiable on $[a, b]$ for the work of this section on Simpson's rule. Define

$$w(x) = \int_a^x (t-a)(t-c)(t-b) dt.$$

It is not hard to show that

$$w(a) = w(b) = 0, \quad w(x) > 0 \quad \text{for } a < x < b.$$

Integrating by parts,

$$\begin{aligned} E_2(f) &= \int_a^b w'(x) f[a, b, c, x] dx \\ &= [w(x) f[a, b, c, x]]_{x=a}^{x=b} - \int_a^b w(x) \frac{d}{dx} f[a, b, c, x] dx \\ &= - \int_a^b w(x) f[a, b, c, x, x] dx. \end{aligned}$$

Applying the integral mean value theorem

$$\begin{aligned} E_2(f) &= -f[a, b, c, \xi, \xi] \int_a^b w(x) dx, \quad \text{for some } a \leq \xi \leq b \\ &= -\frac{f^{(4)}(\eta)}{24} \left[\frac{4}{15} h^5 \right], \quad h = \frac{b-a}{2}, \quad \text{for some } \eta \in [a, b]. \end{aligned}$$

Thus

$$E_2(f) = -\frac{h^5}{90} f^{(4)}(\eta), \quad \eta \in [a, b].$$

Section 4.4 Newton-Cotes formulas

Section 4.5 Romberg Integration

Section 6.1/6.2/6.6 Gaussian Elimination/Solving Linear Systems/Pivoting/Special Matrices (Text only)

Section 7.1 Matrix Algebra Review (Text only)

The Gauss-Jacobi method (Simultaneous displacements). Rewrite $Ax = b$ as

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} x_j \right), \quad i = 1, 2, \dots, n. \quad (16)$$

assuming all $a_{ii} \neq 0$. Define the iteration as

$$x_i^{(m+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} x_j^{(m)} \right), \quad i = 1, \dots, n, \quad m \geq 0. \quad (17)$$

and assume initial guesses $x_i^{(0)}$, $i = 1, \dots, n$, are given. There are other forms to the method. For example, many problems are given naturally in the form

$$(I - B)x = b$$

and then we would usually first consider the iteration

$$x^{(m+1)} = b + Bx^{(m)}, \quad m \geq 0. \quad (18)$$

Our initial error analysis is restricted to (??), but the same ideas can be used for (??). To analyze the convergence, let $e^{(m)} = x - x^{(m)}$, $m \geq 0$. Subtracting (??) from (??),

$$e_i^{(m+1)} = - \sum_{\substack{j=1 \\ j \neq i}}^n \frac{a_{ij}}{a_{ii}} e_j^{(m)}, \quad i = 1, \dots, n, \quad m \geq 0. \quad (19)$$

Hence

$$|e_i^{(m+1)}| \leq \sum_{\substack{j=1 \\ j \neq i}}^n \left| \frac{a_{ij}}{a_{ii}} \right| \|e^{(m)}\|_\infty.$$

Define

$$\mu = \max_{1 \leq i \leq n} \sum_{\substack{j=1 \\ j \neq i}}^n \left| \frac{a_{ij}}{a_{ii}} \right|. \quad (20)$$

Then

$$|e_i^{(m+1)}| \leq \mu \|e^{(m)}\|_\infty$$

and since the right side is independent of i ,

$$\|e^{(m+1)}\|_{\infty} \leq \mu \|e^{(m)}\|_{\infty}. \quad (21)$$

If $\mu < 1$, then $e^{(m)} \rightarrow 0$ as $m \rightarrow \infty$ with a linear rate bounded by μ , and

$$\|e^{(m)}\|_{\infty} \leq \mu^m \|e^{(0)}\|_{\infty}. \quad (22)$$

In order for $\mu < 1$ to be true, the matrix A must be *diagonally dominant*, that is, it must satisfy

$$\sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| < |a_{ii}|, \quad i = 1, 2, \dots, n. \quad (23)$$